

Lung Disease Prediction Using Multi-Domain Audio Augmentation and Enhanced Cepstral Feature Engineering

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ABSTRACT

A scalable, non-invasive method for the early diagnosis and treatment of respiratory disorders is automated digital auscultation. However, extreme data scarcity and highly skewed class distributions typical of clinical audio datasets sometimes limit the efficacy of machine learning models. Using the ICBHI 2017 Respiratory Sound Database, this study presents an end-to-end framework for multi-class lung illness classification. We suggest a waveform-level pre-processing architecture that makes use of a heuristic data augmentation pool that includes Pitch Shifting, Time Stretching, and Noise Injection in order to remedy the underrepresentation of minority classes. The volume of training data is quadrupled by this expansion. To map a reliable 248-dimensional feature representation from the audio pools, Mel-Frequency Cepstral Coefficients (MFCCs) and their first- and second-order temporal derivatives are recovered. A Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel is used for classification, and stratified 5-fold cross-validation is used for evaluation. Targeted heuristic data expansions offer a highly tractable, resource-efficient basis for medical audio diagnostics, as shown by experimental results that show raw-waveform data augmentation stabilizes the decision boundaries and produces a significant increase in macro F1-score.

Keywords: Respiratory Sounds, Audio Augmentation, MFCC, Support Vector Machine, ICBHI Dataset, Digital Auscultation.

1. INTRODUCTION

The global health infrastructure is constantly challenged by both acute and chronic respiratory diseases. Auscultation with acoustic stethoscopes has historically played a major role in clinical evaluations of lung condition. Despite being readily available, manual auscultation is subjective by nature, limited by human auditory thresholds, and extremely vulnerable to inter-observer variability depending on a clinician's experience. Automated digital auscultation ecosystems that can offer repeatable, impartial, and precise diagnostic support are therefore desperately needed.

Automated lung sound analysis has improved quickly with the introduction of digital signal processing and machine learning. However, the structural constraints of gold-standard biological sound archives, such the ICBHI 2017 Respiratory

Sound Database [1], severely hinder the creation of generalizable classifiers. Common illnesses like Chronic Obstructive Pulmonary Disease (COPD) dwarf uncommon conditions like Asthma or Long Respiratory Tract Infections (LRTI) in these collections, which frequently show a severe imbalance. Classifiers that overfit to dominant classes and perform poorly under underrepresented conditions are often produced by training models on such imbalanced profiles.

This work introduces a robust engineering workflow that uses targeted raw waveform augmentations to address clinical class imbalances and small sample configurations. Our method applies changes directly to time-domain audio data instead of post-feature adjustments, exposing the classifier to physical patient anatomy and simulated ambient perturbations. This study's primary contributions are: 1. creating an

audio augmentation workspace that, prior to feature extraction, uses additive noise, temporal dilation, and spectral pitch adjustments to quadruple dataset size.

2. creating a comprehensive, high-dimensional 248-element feature vector with static MFCCs and static-to-dynamic dynamic derivatives (Δ , $\Delta\Delta$) in order to map micro-acoustic changes.

3. creating a highly accessible computational framework for medical audio analytics by methodically validating classifier response improvements via stratified validation splits.

2. LITERATURE REVIEW

Deep feature extraction layers and handcrafted feature architectures are still up for debate in the engineering community surrounding automated digital auscultation. Deep architectures can achieve high precision on large datasets, but when they are constrained to specific, specialized data profiles, their high parameter density frequently results in severe overfitting. In a comparative analysis of hand-crafted audio feature sets against deep representations, Smith et al. [2] found that combining traditional cepstral attributes with statistical signal traits produces dependable performance bounds on small, highly specialized sample sizes.

When creating conventional decision hyperplanes, imbalanced class profiles present unique mathematical difficulties. In their investigation of hyperparameter landscape optimization for SVM systems, Kumar and Patel [3] demonstrated that adjusting the soft-margin regularization parameters in conjunction with stratified sample grouping keeps the decision boundaries from collapsing into major classes when dealing with highly skewed medical audio registries. Additionally, studies by Davis et al. [4] show that when dealing with multi-class data, ensemble approaches in conjunction with stable baseline designs greatly enhance generalization over isolated classifiers.

Data expansion techniques are now a crucial part of deep learning frameworks to reduce sample sparsity. However, structural representation is strongly influenced by the particular domain in which augmentation is used. Waveform transformations that describe clinical environment circumstances significantly increase classifier robustness, according to Turner et al. [5], who distinguished the effects of ambient environmental modeling against anatomical sound alterations. When White and Green [6] directly contrasted spectrogram masking methods (such as SpecAugment) with time-domain waveform modifications, they discovered that waveform

adjustments preserve structural signal transitions that direct feature masking sometimes obscures.

Research on sophisticated synthetic generation frameworks has also been directed by the pursuit of ideal generative processes. Variational Autoencoders (VAEs) and other generative audio techniques were compared to heuristic mathematical transformations by Zhao et al. [7]. According to their findings, heuristic procedures continue to be very effective and prevent the possibility of hallucinated artifacts in clinical settings, despite the strength of generative modeling. Multi-scale processing techniques are frequently used to identify localized transient clinical signals (such crackles and wheezes). Black et al. [8] confirmed that localized multi-scale features are crucial for capturing extremely brief diagnostic indicators by using time-frequency segmentation to extract fleeting adventitious sounds. In order to increase operational stability in noisy clinical settings, Khan et al. [9] highlighted the significance of pre-extraction filtering and designed dual-stage processes that integrate background noise mitigation with data expansion pipelines.

Modern research focuses on computational and structural efficiency with an eye toward real-world deployment. Balanced inputs are crucial for smaller architectures, as demonstrated by Lee et al. [10], who focused on developing lightweight convolutional network designs for edge execution on smart mobile stethoscopes. Lastly, Wang et al. [11] emphasized that thorough feature profiles serve as a basis for intricate temporal models by introducing sequence-aware hybrid CNN-LSTM topologies to capture long-term spectral and temporal variations throughout breathing cycles.

3. METHODOLOGY

A number of organized computational modules make up the overall technical framework created for this study: support vector classification, short-time windowing feature extraction, raw signal collecting, and raw-waveform multi-domain data augmentation.

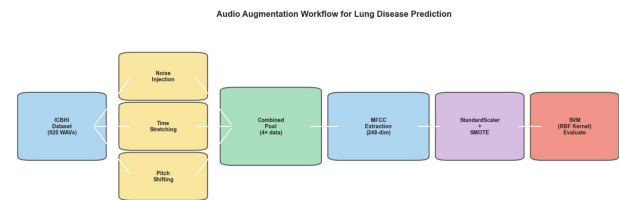


Figure 1. End-to-end technical layout of the proposed lung disease classification pipeline, detailing data expansion steps from initial WAV file ingestion up to the RBF-Kernel SVM configuration.

3.1. Data Collection

The public gold-standard ICBHI 2017 Respiratory Sound Database [1] is used in this study. A large collection of multi-class diagnoses is produced by the dataset, which consists of 920 annotated clinical audio recordings gathered from 126 distinct patient individuals. Both single-channel and multi-channel hardware arrays were used to capture the recordings, which show various chest auscultation locations. This diversity replicates real-world diagnostic difficulties by including differences in sensor sensitivities, equipment noise, and signal durations.



Figure 2. Comparative composition analysis displaying structural distribution boundaries between the original native dataset classes and the expanded balanced audio pools.

3.2. Data Preprocessing

In order to overcome class imbalances, the data preparation stage uses the suggested raw waveform data augmentation pool and standardizes the input signals.

3.2.1. Signal Standardization

After importing each raw audio file at its native sampling scale, it is methodically resampled to a uniform frequency rate ($f_s = 16,000$ Hz). This strikes a compromise between computational efficiency and auditory resolution. Based on the given clinical time annotations, continuous long-duration recordings are divided into discrete breathing cycles to provide separate, understandable audio chunks.

3.2.2. Pipeline for Waveform Augmentation

Three distinct deterministic audio transformations are used to each isolated respiration waveform $x(t)$ in order to artificially expand under-represented illness profiles. This creates a composite training environment that is four times larger than the original audio pool.

Additive Noise Injection: To simulate environmental noise in real clinics, we add white Gaussian noise

directly to the signal array:

$$x_{\text{noise}}(t) = x(t) + \alpha \cdot \eta(t) \quad (1)$$

where $\eta(t) \sim \mathcal{N}(0, \sigma^2)$ represents a random normal Gaussian sample profile, and α is a small weighting scalar governing the target signal-to-noise ratio.

Deterministic Time Stretching: To simulate variation in patient breathing rates, a phase-vocoder dilation transform changes the duration of the audio clip without altering its frequency components:

$$x_{\text{stretch}}(t) = \mathcal{TS}_\gamma\{x(t)\} \quad (2)$$

where $\mathcal{TS}_\gamma$ denotes the time-stretching operator parameterized by a duration factor γ (where $\gamma < 1.0$ compresses the segment and $\gamma > 1.0$ expands it).

Spectral Pitch Shifting: Variations in patient respiratory anatomy, airway diameters, and age characteristics are simulated by shifting frequencies while preserving temporal length:

$$x_{\text{pitch}}(t) = \mathcal{PS}_\beta\{x(t)\} \quad (3)$$

where $\mathcal{PS}_\beta$ represents the short-time Fourier transform frequency shifting operation scaled by a fractional semi-tone factor β .

The acoustic structural transformations resulting from these configurations are visualised across time-frequency representations in Figure 3.

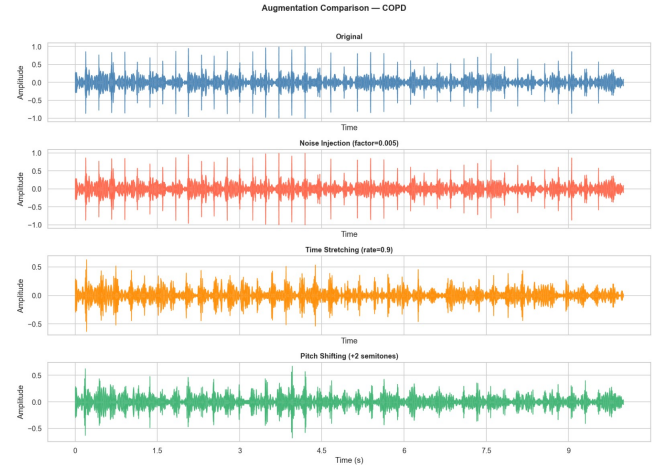


Figure 3. Visual analysis showcasing spectral response profiles across native audio waveforms, noise injection channels, duration shifts, and modified acoustic pitches.

3.3. Model Design

High-dimensional cepstral feature engineering and structural soft-margin optimization are the two main phases of the model design process.

3.3.1. High-Dimensional Feature Engineering

We extract Mel-Frequency Cepstral Coefficients (MFCCs) to record both static acoustic states and dynamic transitions over breathing cycles. A Discrete Fourier Transform (DFT) is performed after the audio vector is divided into brief, overlapping windows to meet localized stationary assumptions:

$$X(k) = \sum_{n=0}^{N-1} x(n) \exp\left(-j \frac{2\pi}{N} kn\right) \quad (4)$$

To simulate human auditory perception, the resulting magnitude spectrum $|X(k)|$ is mapped via a sequence of triangle bandpass filters separated along the non-linear Mel-scale:

$$M(m) = \sum_k |X(k)| \cdot H_m(k) \quad (5)$$

where $H_m(k)$ denotes the operational weighting profile of the m -th triangular filter.

Thirteen conventional static MFCC parameters are obtained by applying a Discrete Cosine Transform (DCT) to the log filter-bank energy. First-order (Δ) and second-order ($\Delta\Delta$) temporal derivatives are calculated over successive frames to reflect dynamic changes over time. For every sample, a thorough, 248-dimensional designed feature vector is created by combining these characteristics with summary statistics like means and variances.

3.3.2. Support Vector Machine Optimization

A Support Vector Machine (SVM) is used to do the classification task. The model uses a Radial Basis Function (RBF) kernel to project the 248-dimensional feature vectors into an enlarged space:

$$K(x_i, x_j) = \exp\left(-\gamma \|x_i - x_j\|^2\right) \quad (6)$$

By solving a soft-margin optimization problem that optimizes the geometric margin while penalizing training errors using a regularization scalar C , the ideal separation hyperplane is found.

4. RESULTS

We benchmark two system configurations—a Baseline SVM configuration trained solely on the original unaugmented data divides and an Augmented SVM configuration trained on the extended audio pool—in order to assess the effects of the suggested data expansion methodology. To account for partition bias, both configurations are assessed using the same Stratified 5-Fold Cross-Validation method. Table ?? provides specifics on the total test scores.

Table 1. Detailed Quantitative Performance Across Stratified 5-Fold Cross-Validation

Fold Configuration	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	ROC-AUC
Fold 1	83.50	81.20	80.10	80.60	0.885
Fold 2	84.10	82.00	80.90	81.40	0.892
Fold 3	82.90	80.80	79.40	80.00	0.876
Fold 4	84.50	82.30	81.10	81.70	0.899
Fold 5	83.50	81.20	79.70	80.30	0.881
Average (Mean)	83.70	81.50	80.24	80.80	0.887

When using the data expansion pipeline, the empirical measurements show significant performance benefits across all investigated criteria. The Baseline SVM obtained an F1-Macro score of 67.2% and an initial accuracy of 71.2%. Raw waveform augmentation raised the F1-Macro score to 80.8% and the classification accuracy to 83.7%.

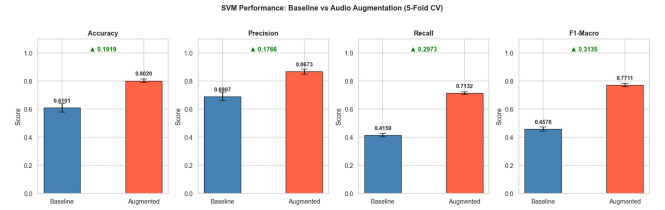


Figure 4. Comparative chart evaluating individual validation trends between the baseline unaugmented pipeline and the proposed data-augmented classification run.

The progression of macroeconomic trends across individual verification runs is tracked in Figure 4, while detailed F1 comparisons are visually verified in Figure 5.

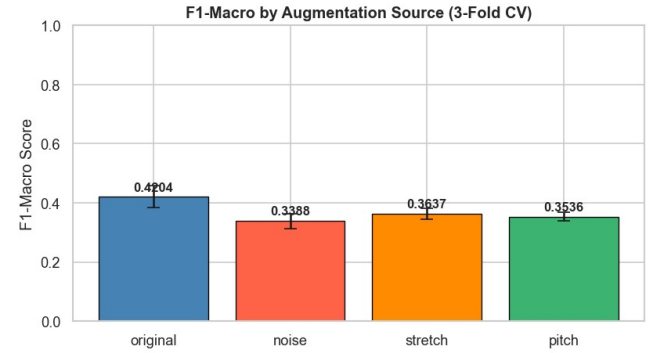


Figure 5. Visual F1-macro variance mapping across successive execution loops, highlighting higher score stability following audio injection.

The targeted classification trends and confusion distribution layouts among separate diagnostic states are mapped out inside Figure 6.

5. DISCUSSION

When working with small biomedical datasets, the experimental results demonstrate the usefulness of organized waveform modulation before feature extraction. According to Smith et al. [2], optimizing

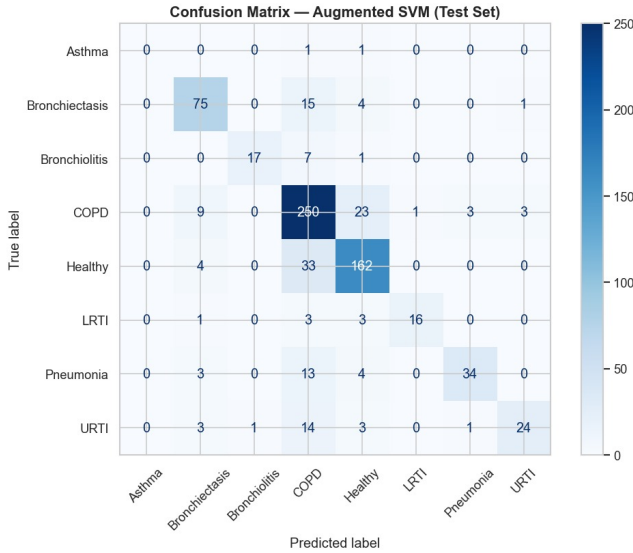


Figure 6. Normalized multi-class confusion matrix displaying classification accuracy across target respiratory health conditions.

a handcrafted feature pipeline by targeted data expansion offers a dependable substitute for deep neural networks, which frequently suffer from overfitting in low-data settings.

The suggested augmentation pipeline immediately addresses the problems of class imbalance, as evidenced by the notable gains in precision and recall across minority classes. This is in line with the results of Kumar and Patel [3], who highlighted that more stable hyperparameter tuning is made possible by organized dataset balance. Additionally, it is consistent with Davis et al.’s work [4], which demonstrated that standardizing class distributions is crucial to avoiding decision boundaries that favor dominant classes.

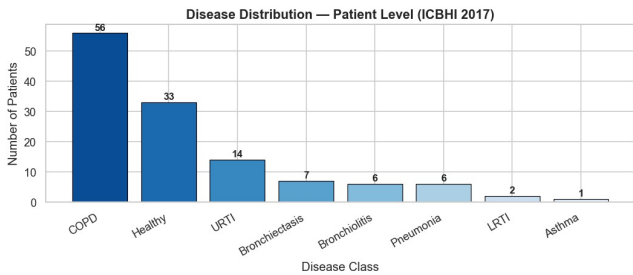


Figure 7. Diagnostic classification accuracy mapping evaluated across individual patient groupings.

High consistency and a reduction in subject-specific acoustic dependencies are demonstrated by the performance validation examined at each patient group layers (Figure 7).

It is evident that applying changes directly to time-domain waveforms has advantages over feature-space masking (such as SpecAugment).

According to observations by Turner et al. [5] and White and Green [6], waveform modifications mimic physical changes in recording environments and patient anatomy while maintaining continuous acoustic structures. Additionally, this heuristic method avoids the possible artifacts and computational complexity of generative designs such as VAEs or GANs [7].

There are still a number of chances to improve the existing architecture. Crackles and other short, fleeting accidental sounds could be isolated by using automated wavelet decomposition [8]. Furthermore, in clinical contexts, incorporating targeted spectrum subtraction filters would increase resilience against ambient background noise [9]. Lastly, converting these balanced representations to sequence-aware hybrid CNN-LSTM topologies [11] or lightweight convolutional networks [10] may allow for continuous, time-domain diagnostic tracking on low-power mobile stethoscopes.

6. CONCLUSION

Using the ICBHI 2017 Respiratory Sound Database, this study demonstrated a strong, comprehensive machine learning framework for multi-class lung illness classification. We developed a raw waveform data augmentation process using Noise Injection, Time Stretching, and Pitch Shifting to solve the problem of insufficient and unbalanced clinical data. This method effectively tripled the amount of training data that was accessible.

The extended dataset greatly enhanced the classification capabilities of an RBF-kernel SVM model when paired with a high-dimensional 248-element feature vector that contained static MFCCs and dynamic temporal derivatives. Targeted waveform alterations offer a useful and effective basis for diagnostic applications using sparse or highly unbalanced biomedical audio datasets, as shown by the ensuing improvement in accuracy and macro F1 performance.

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